

APPLIED ELECTRONICS

Lielina Solomon Getachew - UGR/6634/17

ASSIGNMENT 01 - NUMBER 7

$$I_s = I_1 + I_2$$

Use current-division:

$$I_1 = I_s \frac{G_1}{G_1 + G_2}, \quad I_2 = I_s \frac{G_2}{G_1 + G_2}$$

Conductances.

$$G_1 = \frac{1}{2}, \quad G_2 = \frac{1}{3}$$

$$G_1 + G_2 = \frac{5}{6}$$

Through the $2\ \Omega$ resistor:

$$I_1 = 12 \cdot \frac{\frac{1}{2}}{\frac{5}{6}} = 12 \cdot \frac{1}{2} \cdot \frac{6}{5} = 7.2\text{ A}$$

Through the $3\ \Omega$ resistor:

$$I_2 = 12 \cdot \frac{\frac{1}{3}}{\frac{5}{6}} = 12 \cdot \frac{1}{3} \cdot \frac{6}{5} = 4.8\text{ A}$$

- Current through the $2\ \Omega$ resistor:

$$I_{2\Omega} = 7.2\text{ A}$$

- Current through the $3\ \Omega$ resistor:

$$I_{3\Omega} = 4.8\text{ A}$$

ASSIGNMENT 02 - NUMBER 1

Write the sum as a single sinusoid:

$$i(t) = A \sin \omega t + B \cos \omega t$$

with $A = 56.5685$, $B = 42.4264$.

Amplitude:

$$R = \sqrt{A^2 + B^2} = 70.71064$$

Phase (radians):

$$\phi = \tan^{-1} \left(\frac{B}{A} \right) = 0.64350139 \text{ rad} \approx 36.8699^\circ$$

Thus

$$i(t) = 70.71064 \sin(\omega t + 0.64350139) = 70.71064 \sin(\omega t + 36.8699^\circ)$$

(Alternatively $i(t) = 70.71064 \sin(\omega t + 36.8699^\circ)$.)

ASSIGNMENT 02 - NUMBER 2

Given:

- $R = 8 \Omega$
- $L = 15.915 \text{ mH} = 0.015915 \text{ H}$
- $f = 50 \text{ Hz}$
- $\omega = 2\pi f = 2\pi(50) = 314.159 \text{ rad/s}$

Inductive reactance:

$$X_L = \omega L = 314.159 \times 0.015915 = 5.0 \Omega$$

Impedance of a series RL coil:

$$Z = R + jX_L = 8 + j5 \Omega$$

Magnitude:

$$|Z| = \sqrt{8^2 + 5^2} = \sqrt{89} = 9.434 \Omega$$

Phase angle:

$$\theta = \tan^{-1} \left(\frac{5}{8} \right) = 32.005^\circ$$

$$Z = 8 + j5 \Omega$$

or

$$|Z| = 9.434 \Omega \angle 32.0^\circ$$